

MINI PROJECT
ON
E-COMPLAINT BOX FOR “COLLEGE”

**Submitted in the partial fulfillment of the requirements for the
award of degree of**

BACHELORS IN COMPUTER APPLICATION BCA

To

UTTRANCHAL INSTITUTE OF COMPUTING SCIENCES

UTTARANCHAL UNIVERSITY , DEHRADUN



**UTTARANCHAL
UNIVERSITY
DEHRADUN**

**NAAC
GRADE A+**

UTTARANCHAL UNIVERSITY

Under supervision of
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ACKNOWLEDGEMENT

The most awaited moment of any endeavor is successful completion, but nothing can be done successfully if done alone. Success is the outcome of contribution and consistent help of various persons and we thank those ones who helped us in successful completion of this project. Primarily I would like to thank **Prof. (Dr.) Sonal Sharma, Director USCS** for providing a healthy and encouraging environment to study. I profusely thankful to the Prof.(**Dr.) Sameer Dev Sharma, Head of Department, USCS** and my Project Mentor **Dr. Manoj Aswal**, Assistant Professor. He has been generous enough to provide me an opportunity and accepting my candidature for the most valuable guidance and affordable treatment given to us at every stage to boost my morale.

Ankush Rawat

BCA-II SEM

DECLARATION

I hereby declare that this project work entitled “**E-COMPLAINT BOX**” submitted by **Ankush Rawat** to **Uttaranchal Institute of Computing sciences**. It is an authentic record of my work carried out at “**UTTARANCHAL UNIVERSITY**” under the guidance of **Dr. Manoj Aswal**

Ankush Rawat

BCA-II SEM

CERTIFICATE OF ORIGINALITY



This is to certify that the project **entitled “COLLEGE AND COMPANIES”** by **Ankush Rawat** has been submitted in the partial fulfillment of the requirements for the award of the degree **of BCA-II** from Uttarakhand University, Dehradun. The results embodied in this project have not been submitted to any other University or Institution for the record of any degree.

Under the guidance of:

Dr. Manoj Aswal

Assistant Professor (USCS)

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1. INTRODUCTION

In any educational institution, maintaining open and effective communication between students and the administration is crucial for ensuring a productive and supportive academic environment. However, traditional methods of handling complaints—such as physical complaint boxes, written applications, or verbal reporting—often prove to be inefficient, slow, and lacking in transparency. These methods can discourage students from voicing their concerns due to fear of exposure, delay in response, or lack of follow-up.

To address these issues, the implementation of an E-Complaint Box System offers a modern, digital solution that improves the way complaints are submitted, managed, and resolved within the college community. This system enables students, faculty, and staff to lodge complaints or suggestions online in a structured and secure manner. Each complaint is categorized and directed to the appropriate department, ensuring timely review and action.

The E-Complaint Box System not only provides convenience and accessibility but also promotes accountability and responsiveness within the institution. It enhances transparency by allowing users to track the progress of their complaints and receive updates on the resolution status. Moreover, the digital nature of the system allows for better documentation, analysis of recurring issues, and improvement in administrative practices.

By adopting such a system, the college can create a more inclusive and responsive environment where every voice is heard, and necessary actions are taken efficiently. This initiative aligns with the broader goal of using technology to improve governance, communication, and student satisfaction on campus.

2. OBJECTIVES

The primary aim of developing an E-Complaint Box is to modernize and simplify the complaint submission and resolution process within an educational institution. Traditional complaint systems often suffer from slow processing, lack of transparency, and poor record management. By introducing a digital platform, the institution can create a more reliable, accessible, and accountable system that encourages users to raise concerns without fear or inconvenience. The objectives listed below outline the specific goals that the proposed system intends to achieve.

1. **To provide a convenient digital platform** for students and staff to submit their complaints anytime and from anywhere.
2. **To ensure proper documentation and centralized storage** of all complaints, preventing loss, misplacement, or tampering of records.
3. **To streamline the complaint-handling process** by automatically categorizing complaints into Administrative or Academic sections.
4. **To improve transparency and accountability** by allowing authorities to track, review, prioritize, and respond to complaints systematically.
5. **To reduce response time** by enabling quick communication between users and the concerned department.
6. **To provide status updates and feedback**, informing users whether their complaint received a positive response or needs further investigation.
7. **To maintain confidentiality and data security**, encouraging users to report issues comfortably and without hesitation.
8. **To enhance overall satisfaction and trust** between students and institutional authorities through a structured, responsive, and efficient complaint management workflow.

3. HARDWARE AND SOFTWARE USED

3.1 SOTWARE REQUIREMENT

- Language used- C ++

3.2 HARDWARE REQUIREMENT

- Minimum requirements
 - ▶ Processor :- Intel Core i3 or equivalent
 - ▶ Ram :- 4 GB
 - ▶ Hard Disk :- 100GB disk space available
 - ▶ Operating System :- Windows 10

4. SYSTEM ANALYSIS

System Analysis is a process of studying an activity in order to define its goals or purpose, improving performance and achieving objectives for portability and growth.

4.1 IDENTIFICATION OF NEED

In many educational institutions, the complaint-handling process is still manual, unstructured, and often inconvenient for students and staff. Traditional methods—such as physical complaint boxes, handwritten forms, or verbal reporting—frequently lead to issues like delayed responses, loss of complaints, lack of confidentiality, and limited tracking. Students may hesitate to report problems due to fear of exposure, lack of follow-up, or uncertainty about whether their concerns will be addressed. These challenges highlight the need for a more reliable, transparent, and efficient complaint management system. As institutions grow in size, the number of administrative and academic issues also increases, making it difficult for authorities to manually monitor, prioritize, and resolve complaints in a timely manner. Without a proper system, important issues may go unnoticed, resulting in dissatisfaction, communication gaps, and reduced trust between students and management. To overcome these limitations, there is a clear need for a centralized **E-Complaint Box** that allows users to submit complaints digitally from anywhere, at any time. A digital system ensures proper documentation, faster communication, and secure storage of records. It also allows authorities to systematically review, categorize, and respond to complaints, improving accountability and efficiency. By providing features such as automated routing of complaints, status tracking, and feedback mechanisms, the proposed E-Complaint Box fulfills the growing requirement for a modern, userfriendly, and transparent platform that supports a healthier and more responsive institutional environment.

4.1.1 EXISTING SYSTEM

Centralized Public Grievance Redress and Monitoring System (CPGRAMS) – India

CPGRAMS is an online platform developed by the **Government of India** for citizens to lodge complaints related to public services. It is one of the most well-known and widely used **e-complaint systems** at the national level.

4.1.2 PROPOSED SYSTEM

The proposed system introduces a digital **E-Complaint Box**, designed to simplify, secure, and modernize the process of registering and resolving complaints within an educational institution. Unlike the traditional manual system, which is time-consuming, unorganized, and lacks transparency, the proposed system provides an efficient and user-friendly online platform for students and staff. In this system, users can log in and submit their complaints by providing essential details such as their name, enrollment number, category of complaint (Administrative or Academic), and the description of the issue. The system ensures that all complaints are stored in a centralized database, enabling easier tracking, retrieval, and management. Once a complaint is submitted, it is automatically forwarded to the concerned authority based on its category.

The proposed system also includes an **admin panel**, where administrators can view all submitted complaints, categorize them, update their status, and provide responses. The system further enhances transparency by allowing users to receive feedback, such as whether the response is positive or negative. If the response is negative, the system displays “Further investigation needed,” and if it is positive, it shows “Thank you, we will get back to you shortly.”

The system ensures data security, reduces paperwork, and eliminates the chances of complaints being lost or ignored. It improves communication between students and authorities by offering a structured way of filing and resolving issues. Overall, the proposed E-Complaint Box is a faster, more reliable, and systematic approach to complaint management that enhances accountability and improves user satisfaction within the institution.

5. PRELIMINARY INVESTIGATION

The purpose of preliminary investigation is to determine whether the problem or deficiency in the current system really exists. It is an inquiry done for a project to determine what the project's requirements are before creating it. For example, for the present system, we need algorithms API'S. The most important requirement for the project was to determine whether the platform we used was appropriate for the project.

5.1 FEASIBILITY STUDY

The purpose of a feasibility study is to determine the viability of a proposed project. Feasibility study is carried out for a number of purposes, including establishing if a software is suitable for development, installation, and business contribution.

Types are :-

TECHNICAL FEASIBILITY

Given the foregoing, all organizations have self-acting, repetitive and time-consuming jobs that humans do. The areas of the existing system that are well suited that makes the project technically feasible.

OPERATIONAL FEASIBILITY

Operational feasibility of a proposed model is assessed by how well it resolves challenges and capitalizes on possibilities identified during project scope, as well as how well it fits the criteria established during the needs analysis phase time period of system development. The present system has automated the bulk of manual activities. As a result, the proposed system will increase admin and user operational efficiency

ECONOMIC FEASIBILITY:

Economic feasibility study is used to show the net benefit of a proposed project for collecting or dispersing electronic money and benefits, while taking into account the company's advantages and expenses.

A thorough assessment of project's development expenses is undertaken as part of this feasibility study, which covers all final development costs, such as hardware and software components, system design, development costs & operational costs. All of the hardware and software technologies utilized in the project's development are freely available and open source. As a result, the costs of development and maintenance are fairly manageable.

6. PROJECT PLANNING

Project planning is a crucial phase in the project development lifecycle, where a comprehensive roadmap is prepared to guide the successful execution of the project. It begins with a clear understanding of the project goals, scope, and expected outcomes. Based on these, the project team identifies all the major tasks and sub-tasks required to complete the project, along with the sequence in which they should be performed.

During project planning, detailed schedules are created using tools such as Gantt charts, timelines, or work breakdown structures (WBS). These tools help in dividing the entire project into manageable phases and assigning deadlines for each activity. Resource planning is also an essential component, where the required manpower, software tools, hardware components, budget, and other materials are carefully assessed and allocated.

Another key part of project planning is **risk management**. Possible risks—such as technical issues, delays, dependency failures, or resource shortages—are identified in advance, and preventive strategies or backup plans are prepared to minimize their impact. Communication planning is also established to ensure that project updates, progress reports, and feedback are shared regularly among team members and stakeholders.

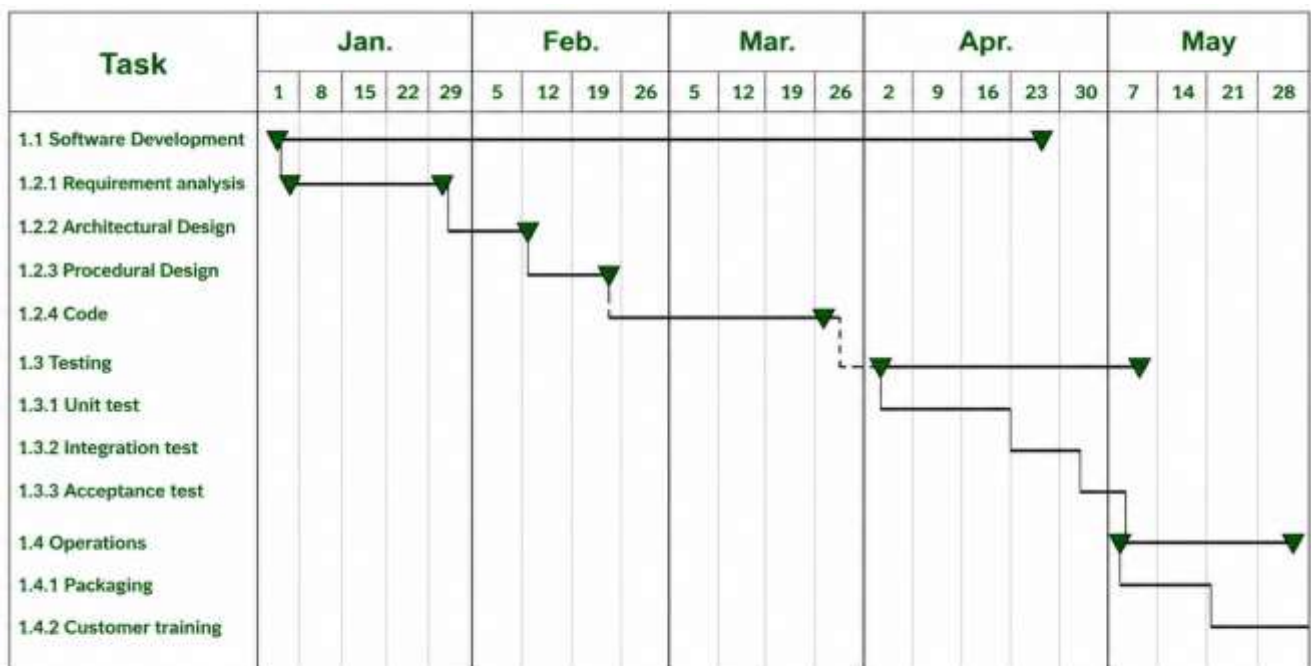
Project planning also includes defining quality standards, testing strategies, documentation requirements, and performance indicators to ensure that the final product meets the intended expectations. By creating a structured plan, the project team can avoid confusion, stay organized, and maintain steady progress throughout the development cycle.

7. PROJECT SCHEDULING

A project schedule is simply a method for conveying and understanding what activities are required & must be accomplished, what organizational resources will be provided or allocated to these tasks, and when the work must be completed. Cost savings, Successful project, & customer pleasure all benefit from effective project scheduling.

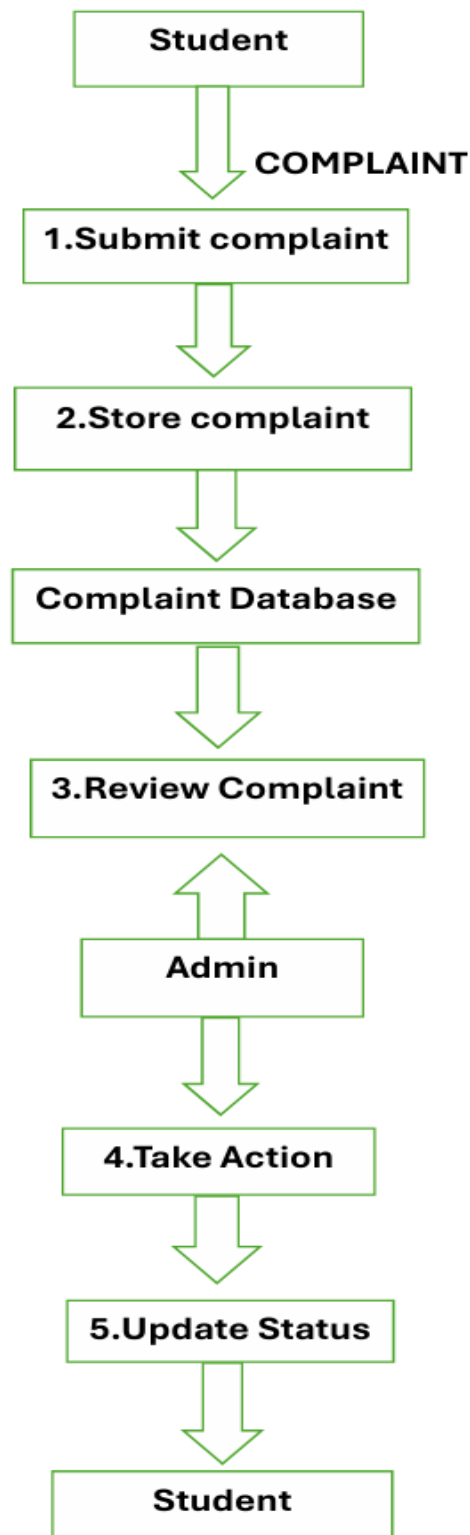
7.1 GANTT CHART

A Gantt chart is a graphical depiction of activities as they relate to time. It is used in project management to assist with project scheduling of all complex programming.



Gantt Chart

8. DATA MODELS



8.0Data flow diagram

8.1 DFD (Data Flow Diagram)

A data flow diagram (DFD) is a graphical or visual representation of a business's operations through data movement that utilizes a recognized set of symbols and notations.

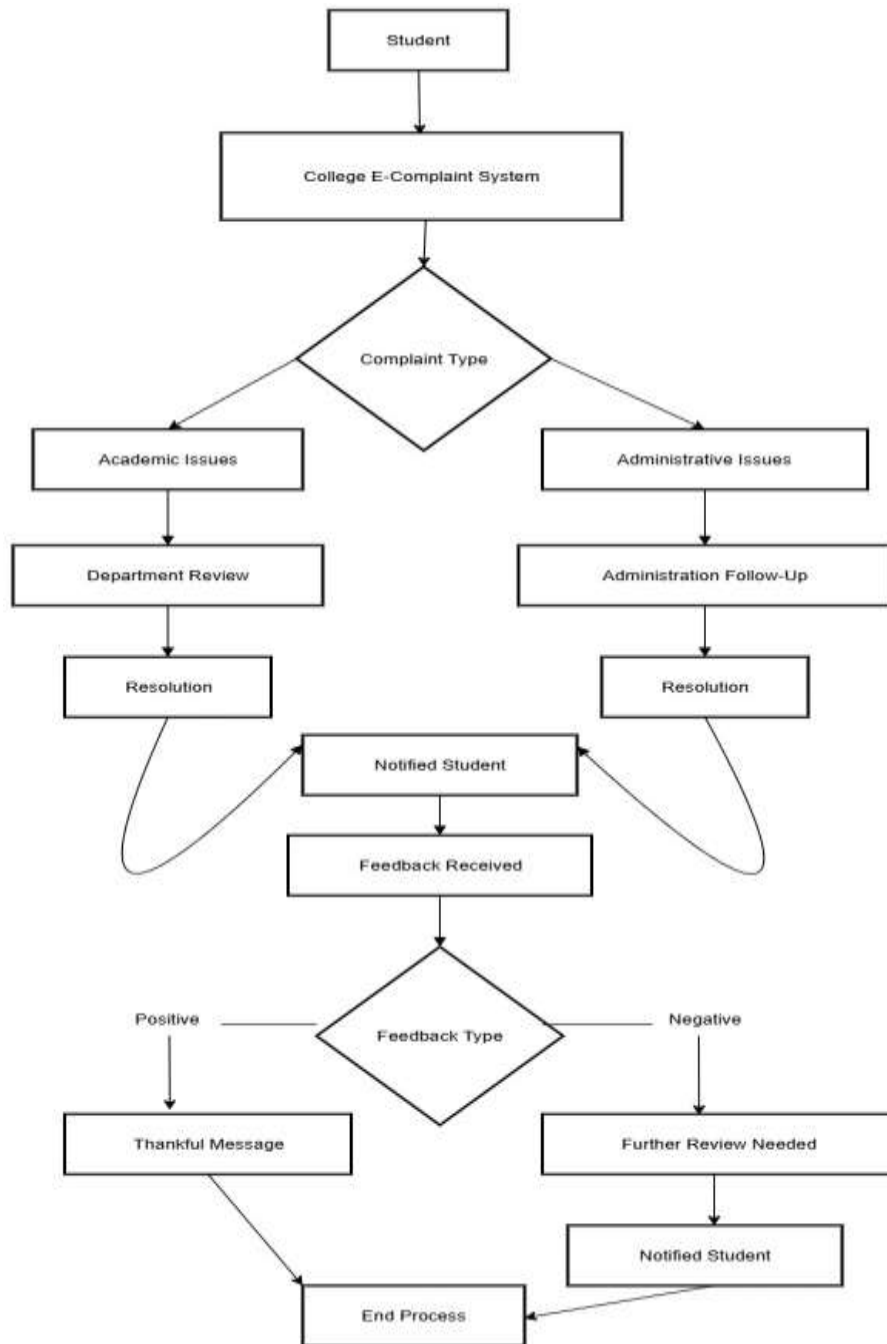


Fig-8.1 Data Flow Diagram

8.2 SEQUENCE DIAGRAM

Sequence diagram is a diagram generated in the Unified Modelling Language (UML) that depicts the sequence of messages shared between objects in an interaction. A sequence diagram is made up of a set of objects represented by lifelines and the messages that they exchange over time during interaction.

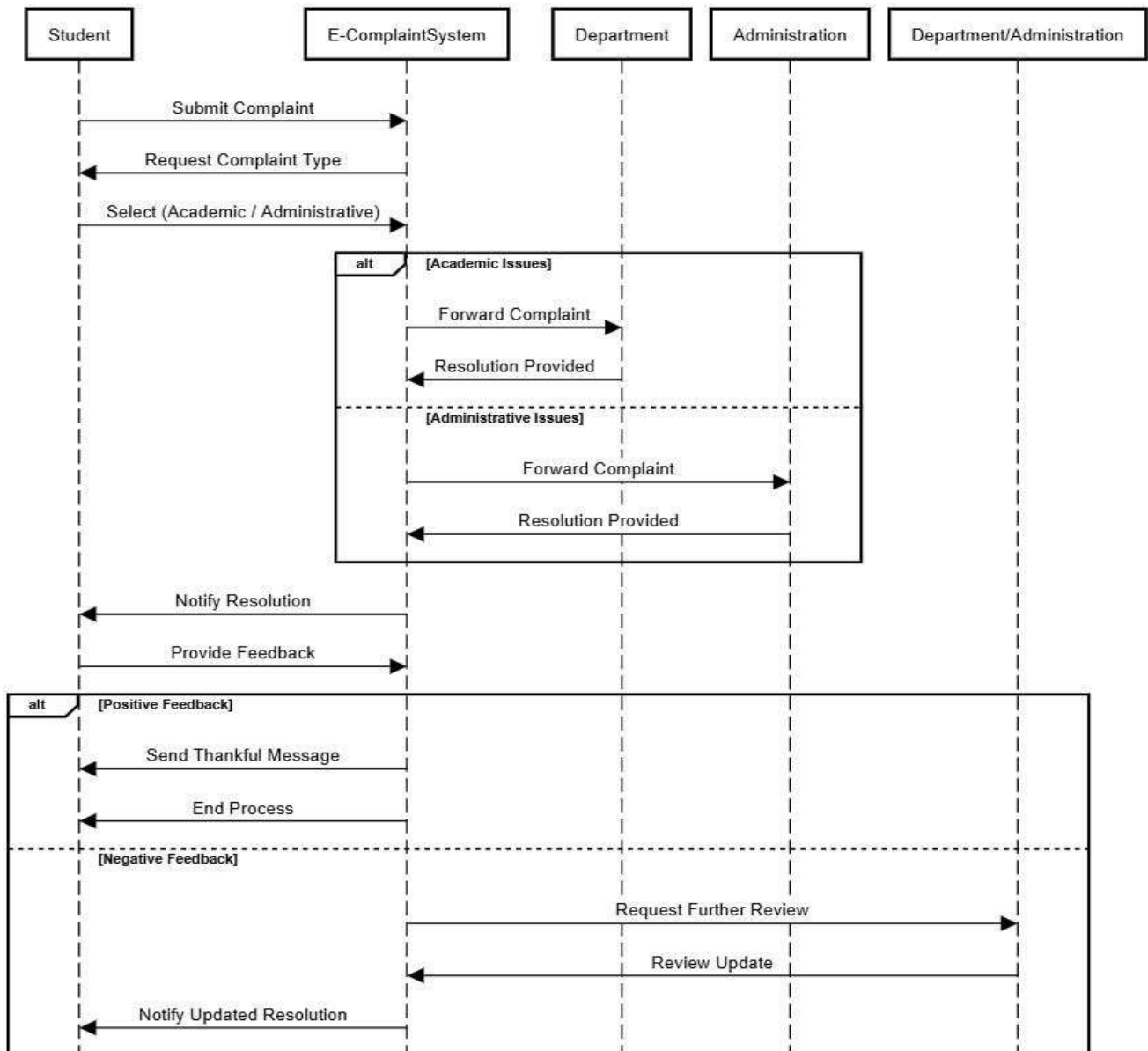
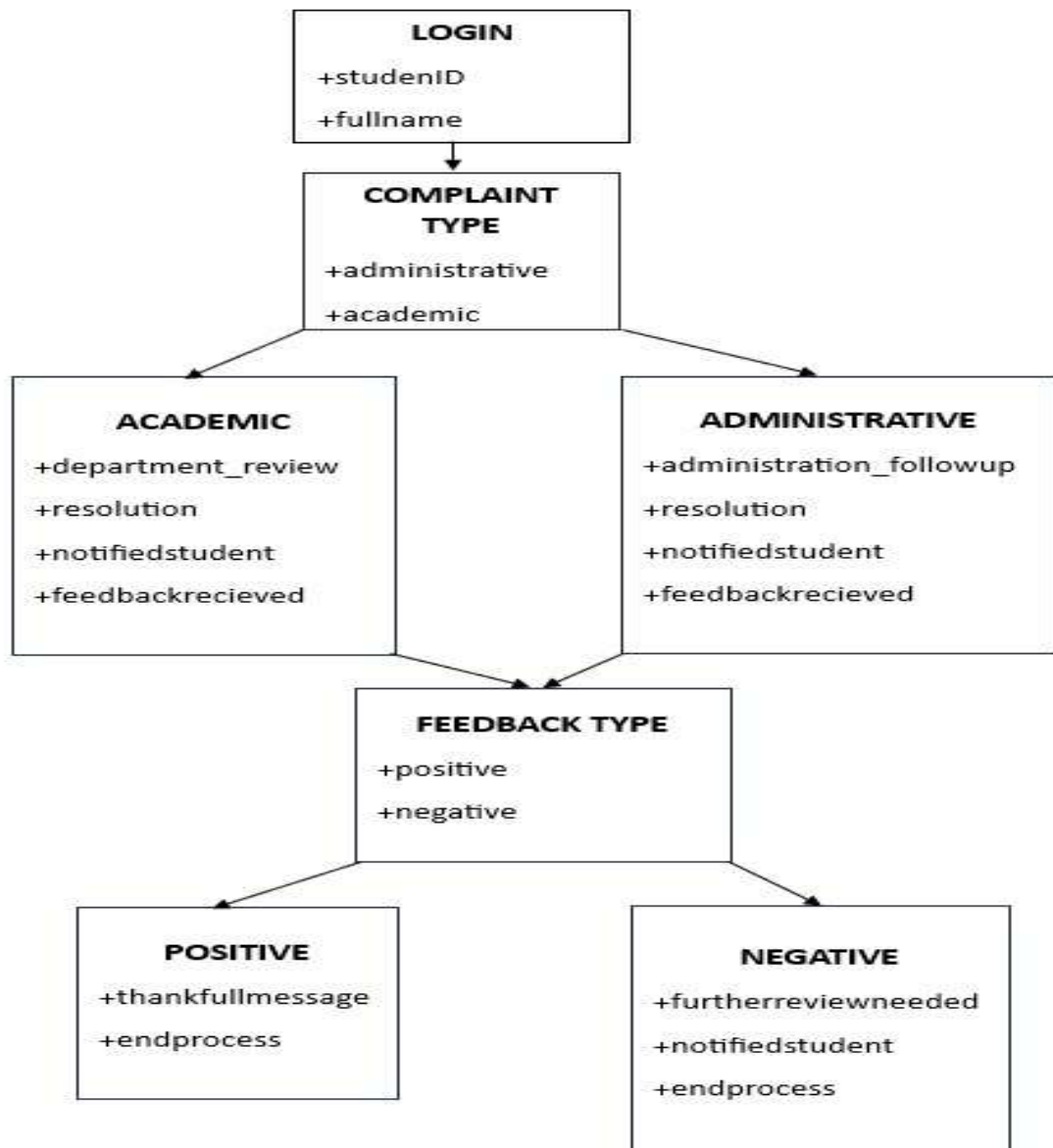


Fig-8.2 Sequence Diagram

8.3 CLASS DIAGRAM

A stable view of a program is depicted in the class diagram. It reflects the many sorts of items in the system as well as their relationships. A class is made up of objects, and it may inherit from other classes. A class diagram is used to illustrate, explain, and record numerous parts of a system, as well as to create executable software code.

It displays the software system's properties, classes, functions, and relationships to provide an overview. It organizes class names, characteristics, and functions into a discrete compartment to aid in programme development. A structural diagram is a collection of classes, interfaces, linkages, collaborations, and constraints.



8.4 ACTIVITY DIAGRAM

Another important behavioral diagram in UML diagram is the activity diagram, which is used to explain dynamic characteristics of the system. An activity diagram is simply a more complex version of a flow chart that depicts the flow of information from one action to the next.

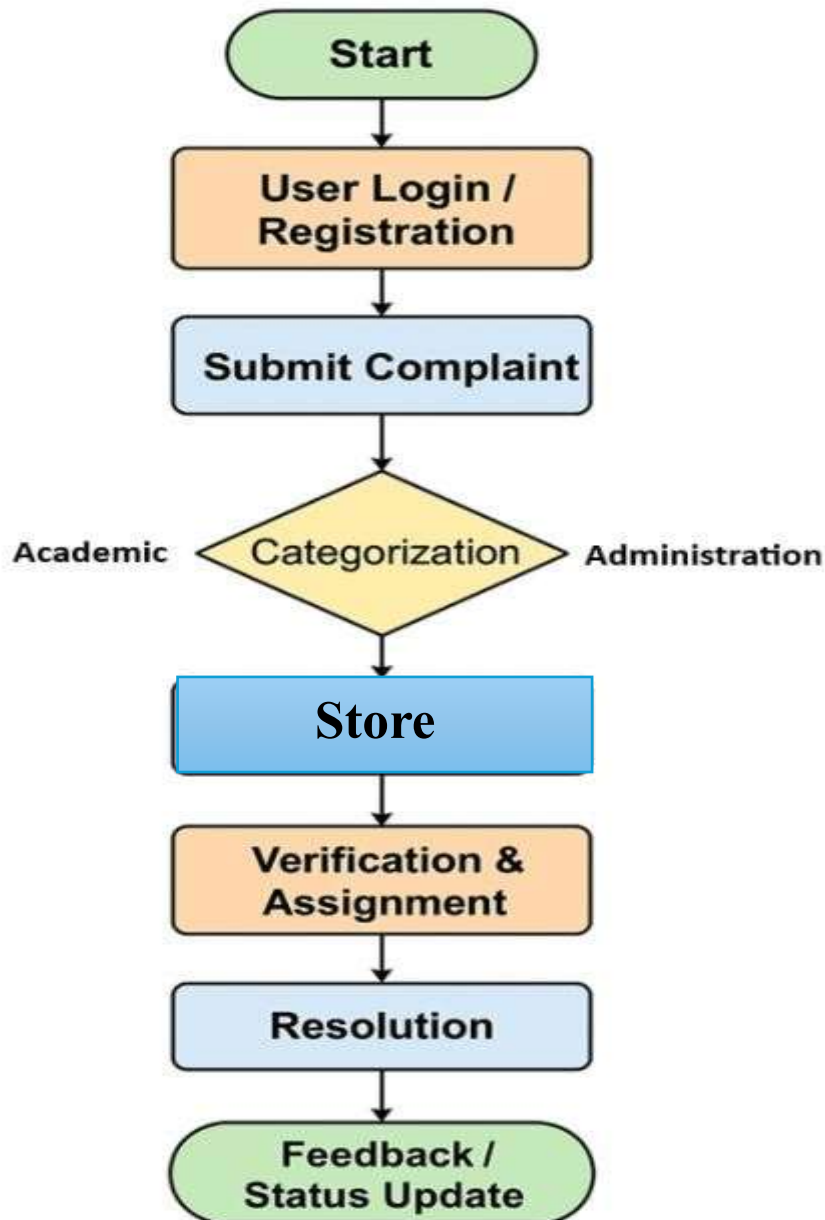


Fig-8.4 Activity Diagram

9. METHODOLOGY

Problem Identification – Recognize the need for a digital complaint system to improve grievance handling.

Requirement Analysis – Gather user requirements (students/employees/citizens) and define system objectives.

System Planning – Decide features such as anonymity, complaint tracking, categorization, and notifications.

Design Phase – Create database design, workflow diagrams, and user interface layout.

Technology Selection – Choose suitable frontend, backend, and database technologies.

Development (Implementation) – Build modules for complaint submission, forwarding, status updates, and dashboards.

Integration – Connect all modules (user, department, admin) for smooth data flow.

Testing – Perform unit, integration, security, and usability testing.

Evaluation & Maintenance – Monitor performance, gather feedback, fix bugs, and update features regularly.

The E-Complaint Box is a digital grievance redressal system designed to collect, manage, and resolve complaints efficiently. Its methodology involves identifying the need for a secure and transparent platform, analyzing user requirements, and planning system features such as complaint submission, tracking, and categorization. The system is then designed with proper database structure, user interfaces, and role-based access. After selecting suitable technologies, the platform is developed, tested for functionality and security, and deployed for real-time use. Continuous evaluation, user feedback, and maintenance ensure that the system remains effective, user-friendly, and updated over time.

E-Complainit Box (Code)

```
#include <iostream>
#include <vector>
#include <fstream>
using namespace std;
struct Complaint {
    string name;
    string enrollment;
    string department;
    string category;
    string problem;
    string status;
};
void showMenu() {
    cout << "\n===== COLLEGE E-COMPLAINT BOX =====\n";
    cout << "1. Submit Complaint\n";
    cout << "2. View All Complaints\n";
    cout << "3. Search by Enrollment Number\n";
    cout << "4. Exit\n";
    cout << "Enter your choice: ";
}
void submitComplaint(vector<Complaint> &complaints) {

    Complaint c;
    cin.ignore();

    cout << "\nEnter Student Name: ";
    getline(cin, c.name);

    cout << "Enter Enrollment Number: ";
    getline(cin, c.enrollment);
```

```

cout << "Enter Department: ";
getline(cin, c.department);

cout << "\nSelect Complaint Category\n";
cout << "1. Academic Problem\n";
cout << "2. Infrastructure Issue\n";
cout << "3. Faculty Issue\n";
cout << "4. Library Issue\n";
cout << "5. Department Issue\n";
cout << "6. Other\n";
cout << "Enter choice: ";

int catChoice;
cin >> catChoice;
cin.ignore();

switch(catChoice) {
    case 1: c.category = "Academic Problem"; break;
    case 2: c.category = "Infrastructure Issue"; break;
    case 3: c.category = "Faculty Issue"; break;
    case 4: c.category = "Library Issue"; break;
    case 5: c.category = "Department Issue"; break;
    case 6: c.category = "Other"; break;
    default: c.category = "Unknown"; break;
}

cout << "\nEnter Your Problem: ";
getline(cin, c.problem);

c.status = "Pending";
complaints.push_back(c);
ofstream file("complaints.txt", ios::app);
file << c.name << "|"
    << c.enrollment << "|"

```

```

        << c.department << "|"
        << c.category << "|"
        << c.problem << "|"
        << c.status << endl;
file.close();

cout << "\nComplaint Submitted Successfully!\n";
}

void viewComplaints(vector<Complaint> &complaints) {

    if(complaints.empty()) {
        cout << "\nNo complaints available.\n";
        return;
    }

    cout << "\n===== ALL COMPLAINTS =====\n";

    for(int i=0;i<complaints.size();i++) {

        cout << "\nComplaint ID: " << i+1 << endl;
        cout << "Name: " << complaints[i].name << endl;
        cout << "Enrollment: " << complaints[i].enrollment << endl;
        cout << "Department: " << complaints[i].department << endl;
        cout << "Category: " << complaints[i].category << endl;
        cout << "Problem: " << complaints[i].problem << endl;
        cout << "Status: " << complaints[i].status << endl;
        cout << "-----\n";
    }
}

void searchComplaint(vector<Complaint> &complaints) {

    string enroll;
    cout << "\nEnter Enrollment Number: ";
    cin >> enroll;

```

```

bool found=false;

for(int i=0;i<complaints.size();i++) {

    if(complaints[i].enrollment==enroll) {

        cout << "\nComplaint Found\n";
        cout << "Name: " << complaints[i].name << endl;
        cout << "Department: " << complaints[i].department << endl;
        cout << "Category: " << complaints[i].category << endl;
        cout << "Problem: " << complaints[i].problem << endl;
        cout << "Status: " << complaints[i].status << endl;

        found=true;
    }
}

if(!found)
    cout << "\nNo complaint found for this Enrollment Number\n";
}

int main() {

    vector<Complaint> complaints;
    int choice;

    do {

        showMenu();
        cin >> choice;

        switch(choice) {

            case 1:

```

```
        submitComplaint(complaints);
        break;

    case 2:
        viewComplaints(complaints);
        break;

    case 3:
        searchComplaint(complaints);
        break;

    case 4:
        cout << "\nExiting System...\n";
        break;

    default:
        cout << "\nInvalid Choice\n";
    }

} while(choice!=4);

return 0;
}
```


Output

Output

```
===== COLLEGE E-COMPLAINT BOX =====
1. Submit Complaint
2. View All Complaints
3. Search by Enrollment Number
4. Exit
Enter your choice: 1

Enter Student Name: Ankush Rawat
Enter Enrollment Number: 25360900101
Enter Department: uscs

Select Complaint Category
1. Academic Problem
2. Infrastructure Issue
3. Faculty Issue
4. Library Issue
5. Department Issue
6. Other
Enter choice: 1

Enter Your Problem: xyz

Complaint Submitted Successfully!

===== COLLEGE E-COMPLAINT BOX =====
1. Submit Complaint
2. View All Complaints
3. Search by Enrollment Number
4. Exit
Enter your choice:
```

10.FUTURE SCOPE

1. In the future, E-Compliant Boxes will use smart technology like sensors to track where they are and check conditions like temperature or damage during shipping.
2. They'll help companies follow rules more easily by storing important info digitally, which can be checked quickly and automatically.
3. These boxes will be made from eco-friendly materials and help reduce waste, making them better for the environment.
4. They might use blockchain (a secure digital system) to keep records of what's inside and where it's been, so nothing gets faked or changed.
5. As online shopping and global trade grow, these smart boxes will make shipping faster, safer, and more organized for everyone.